

GOKU TEAM · AGENTIC AI BUILD WEEK 2026

KFC Ordering Agent

KFC customers are already messaging on Messenger. We turn the conversation into a *completed order* — with an AI agent that plans, acts through tools, and verifies itself.

F&B

AI Engineering

Product

Problem 4 — AI-powered conversational ordering via chat

Customers are already in chat. The operator is the ceiling.

WHO

High-intent customers who message the fanpage before opening the app

GOAL

Finish the order inside the chat, no channel switch

FRICTION

Ordering means leaving chat for the app — conversion leaks midway

ROOT CAUSE

100% handled by hand — the operator's speed caps chat revenue

EVIDENCE

KFC brief: 250+ stores, millions of customers a month, no conversational ordering

ONE INSIGHT

KFC's chat channel isn't short on customers — it's short on **someone to close orders 24/7.**

The agent doesn't create new demand. It harvests demand that leaks away every day.

Goal → Plan → Tools → Act → Verify, on one real message

Customer: “deliver to 350 Tran Hung Dao, D5, phone 090..., pay cash please”

GOAL

Understand — close a COD order to a District 5 address



PLAN

Resolve store → check sold-out items → close COD



TOOLS

set_delivery_info ·
select_payment_method (2 of 14 tools)



ACT

Route to KFC Nguyen Trai (open to 22:00) · create order KFC-0001



VERIFY

State machine + service validate, two layers — wrong step = tool locked

NOT A BLACK BOX

The staff console shows the **decision log each turn**:  Save address · resolve store (KFC Nguyen Trai) → Create order & pay (KFC-0001 · cash) — judges watch the agent decide in real time.

TOOL DISCIPLINE

14 tools locked across 11 order states — the payment tool doesn't exist to the LLM until an order is confirmed. Saying “ordered” requires a real order code from a tool.

The LLM holds the conversation. The state machine holds the money.

Hybrid agent + state machine

The LLM never decides price or order on its own — every number comes from the DB through tools, validated twice.

→ Orders close in under 2 minutes, never a made-up price.

Store-aware, 250 stores

Orders route to the right store by address, knowing opening hours and sold-out items before accepting.

→ Never accepts an order it can't deliver — operations you can trust.

Humans in the right place

Complaints and sensitive cases hand off; staff take over in one tap, and the customer always knows who they're talking to.

→ Controlled automation — enterprise-grade.

Why can't a general assistant do this? Because it isn't allowed to handle money. This system is — safely.

What's measured, what's assumed — labeled honestly

LEVEL 2 · BENCHMARK — MEASURED

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Vietnamese NLU auto-eval with trap cases against fabricated items · re-run after every change (caught 2 regressions in a day)

LEVEL 1 · ASSUMPTION — LABELED

+2.7%

AOV uplift, backtested on 4,964 simulated POS transactions at 20% acceptance — conservative vs. KFC's 10–15% brief estimate

IMPACT AT SCALE

Chat orders/mo × AOV × uplift

+ recovered chat conversion (now 0, hand-handled) + reduced operator load · pilot measures via A/B

11-step

Order completion flow, running live

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NLU accuracy · gpt-5.4

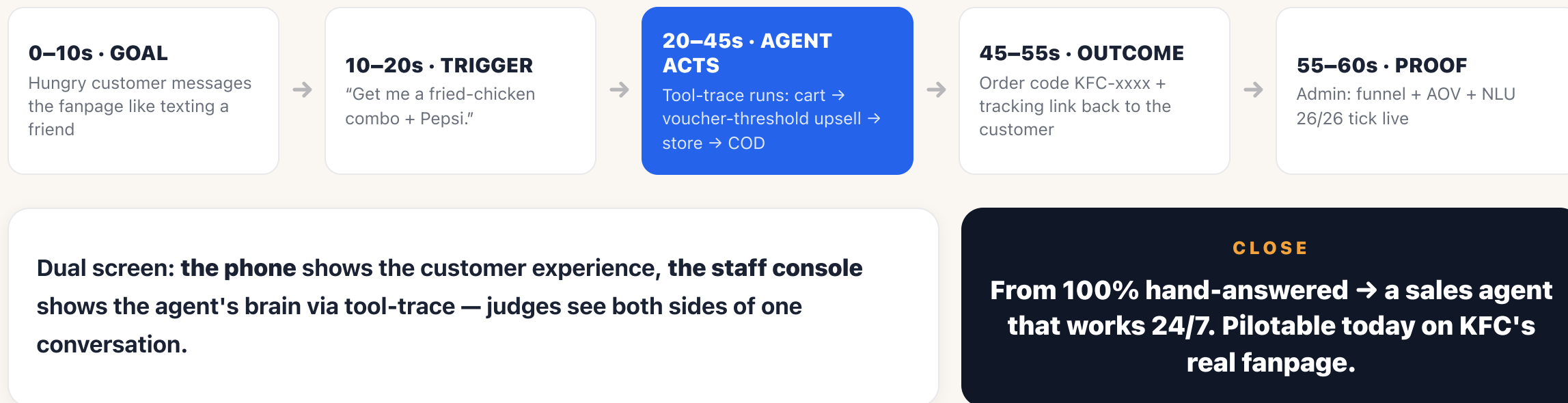
Threshold

Upsell: "+11k → free ship, save 20k"

Auto

Best voucher applied, nothing to memorize

60-second demo — ending on the customer's outcome



KFC ORDERING AGENT · GOKU TEAM

Thanks for watching!

"Text KFC like texting a friend — and the order runs itself to your door."

kfc-ordering-agent.vercel.app

APPENDIX · A1 – A8

Evidence for Q&A

We answer first — and show an appendix slide only when a claim needs support.

An 11-state machine drives the order lifecycle

BROWSING → CART → CONFIRMING → COLLECTING_DELIVERY → SELECTING_PAYMENT

└─ COD → **PLACED** → PREPARING → DELIVERING → DELIVERED

└─ QR/card → AWAITING_PAYMENT → **PLACED** → ...

TOOLS LOCKED BY STATE

Each state opens only a subset of the 14 tools — the LLM can't check out before an order is confirmed, even under prompt injection.

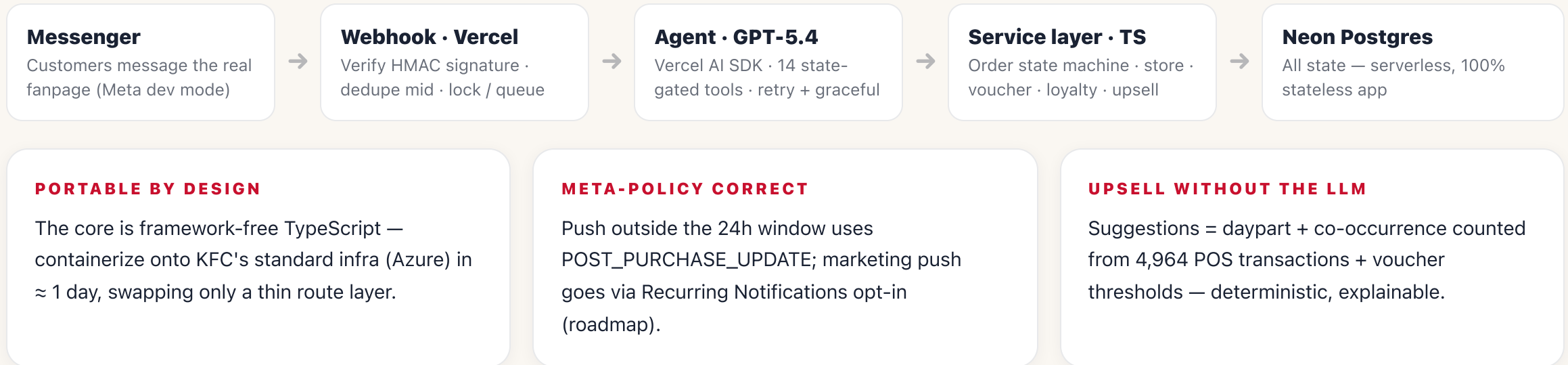
TWO-WAY SYNC

Staff change the order state → the agent session syncs → the customer gets a push; a drifted session self-heals on the next turn.

MANY CUSTOMERS AT ONCE

Per-conversation lock (CAS) + queue + dedupe by mid + consecutive messages merged into one LLM turn.

Architecture is evidence — not the main character



Automation with brakes

Money never touches the LLM

Price, total and voucher are computed only by the service from the DB. Tools validate an item exists before adding to cart. Saying “ordered” requires a real order code from a tool.

Human oversight

Complaints and sensitive cases (allergy, refund) hand off. Staff take over in one tap — the bot goes silent instantly, and the customer is told clearly whether they're talking to a person or a bot.

Customer data

Phone numbers masked in admin and logs · webhook verifies signatures and rejects unsigned requests in production · secrets in env · public privacy policy.

Fault tolerance

LLM error → retry, then apologize gracefully; messages are never lost (peek / mark-after) · idempotent payments · the /pay page blocks already-processed orders.

Tested for real, found real bugs, fixed same-day

OBSERVED

After delivery, the customer's next order is rejected

ROOT CAUSE

Session stuck in the old state + cart not cleared after the order

FIX + VERIFY

Self-healing session + two-way state sync + cart clear — smoke test 6/6

OBSERVED

Bot says "I'll order now" but creates no order

ROOT CAUSE

LLM "says but doesn't do" — missing a say=do constraint

FIX + VERIFY

SAY=DO rule + absolute priority for the order tool — reproducing eval case passes

OBSERVED

Bot makes up a combo price from old chat history

ROOT CAUSE

In a stuck state the agent loses tools, so it acts in words

FIX + VERIFY

TOOL > HISTORY in the prompt + enough tools open in every valid state

Every bug becomes a case in the eval suite — the system never repeats an old error.

Formula first, numbers second

Value / year = chat orders per month × AOV × uplift × 12 + recovered chat conversion + page-operator hours

MEASURED (MOCK, TRANSPARENT)

Backtest of 4,964 POS transactions: AOV +2.7% at 20% acceptance · voucher-threshold upsell lifts basket value right in the demo.

FROM THE KFC BRIEF

KFC estimates context-right upsell lifts AOV 10–15% · the chat channel currently converts ≈ 0 because 100% is hand-handled.

PILOT MEASURES FOR REAL

A/B on the real fanpage: completion rate, AOV delta, response time, CSAT — 2 weeks of real numbers replace every assumption.

LLM cost is under 2% of order value — and only the chat channel uses the LLM; the upsell engine runs deterministically, for free.

Compared with two familiar paths

Rule-based bot (button menus)

Can't parse "2 spicy chickens, mild, deliver D5, pay cash" in one sentence · every flow change means recoding the button tree

✗ **Rigid, unnatural**

General LLM assistant

Answers well but isn't allowed to handle money: no state machine, no tool gating → made-up prices and mis-orders are a certainty

✗ **Uncontrolled**

Our hybrid

The LLM handles natural language · a state machine + 14 step-locked tools handle money and orders · handoff handles exceptions

✓ **Natural + safe**

"GenAI where language is needed — deterministic where money is touched."

Automated eval — we don't trust vibes

26/26

Vietnamese NLU eval · gpt-5.4

Run through the real agent + real services + real DB

NLU (9 cases)

natural item ordering, quantity, ambiguous size → re-ask, multiple intents in one sentence

Flow (8 cases)

confirm → delivery → COD / QR → tracking → cancel, all by the rules

Guardrail (6 cases)

nonexistent item → no fabrication · early COD mention → still creates a real order

Store-aware (3 cases)

resolve the right district · store closed → switch · sold out → suggest an alternative

The eval is a regression brake: it once dropped to 21/24 after a change — caught and fixed BEFORE deploy.

Knowing exactly where we stand

CURRENT LIMITS (DEMO)

- Synthetic data — the brief confirmed the direction is right absent real data
- Transparent mock payments — no real gateway connected yet
- Stores matched by district, no delivery-radius check yet
- Messenger dev mode — users need a tester role

ROADMAP AFTER PILOT

- Connect KFC's real POS / OMS + loyalty (schema already mirrored)
- VNPay / MoMo webhook · distance API · per-store menu overlay
- Zalo on the same agent core (just add an adapter) · P2 kiosk on the same upsell engine
- Recurring Notifications opt-in · anti-fraud (COD limit, blacklist)

Q&A — Answer → Support → Connect

1 · ANSWER

Conclusion first, no hedging.

"Yes — the agent handles many customers at once."

2 · SUPPORT

Exactly one fact.

"Per-conversation lock + queue, tested with two phones in parallel."

3 · CONNECT

Back to user value.

"So at the lunch peak, no customer waits."

Asked something we don't have: "Not validated yet — what we know is X, and we'll test it with Y in the pilot." · Opening Q&A: "May I gather the questions first?" (0:20 gather · 1:20 answer)

ONE LINE PER RUBRIC CRITERION — THE WHOLE TEAM KNOWS THESE BY HEART

Six one-line answers, one per rubric criterion

AGENTIC AI

The agent plans and acts through 14 state-locked tools — the tool-trace shows each decision in real time.

TRACK FIT

Solves KFC's P4 exactly: the chat channel is 100% hand-handled; we turn it into the first automated sales channel.

EXECUTION

An 11-step flow live on Messenger: order → pay → track → deliver — eval 26/26, smoke 25/25.

IMPACT

Orders close in chat in under 2 minutes, voucher-threshold upsell lifts AOV, the page team is freed from repetitive work.

CREATIVITY

Hybrid: the LLM holds the conversation, the state machine holds the money — a general assistant can't handle money; this system can, safely.

CLARITY

One sentence to retell: "Text KFC like texting a friend — and the order runs itself to your door."